

# HAMZA ASIM

AI Engineer | Generative AI | Machine Learning | Agentic Systems

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## PROFESSIONAL SUMMARY

AI Engineer and Data Science graduate with hands-on experience building production-oriented generative AI, agentic automation, NLP and machine learning systems. Skilled in Python, LLM/RAG workflows, MCP, vector search, FastAPI, data pipelines and cloud deployment across AWS, Azure and GCP. Experienced delivering end-to-end AI solutions from data ingestion and model integration to APIs, orchestration, evaluation and deployment, with measurable gains in accuracy, retrieval speed and operational efficiency.

## CORE TECHNICAL SKILLS

**Generative AI & Agents:** Large Language Models (LLMs), RAG, AI Agents, Multi-Agent Systems, Model Context Protocol (MCP), MCP Servers, Tool/Function Calling, Prompt Engineering

**Retrieval & LLM Systems:** Embeddings, Semantic Search, Vector Databases, Qdrant, Context/State Management, Structured Outputs, LLM Orchestration

**Machine Learning & NLP:** Python, Scikit-learn, Pandas, NumPy, PySpark, NLP, Classification, Information Extraction, Feature Engineering, Model Evaluation

**Backend & Data Engineering:** FastAPI, REST APIs, PostgreSQL, Redis, Snowflake, Apache Airflow, SQL, ETL/Data Pipelines, Microservices

**Cloud & DevOps:** Docker, Git/GitHub, Linux, AWS, Azure, GCP, AWS RDS, API Integration, Model Deployment

## PROFESSIONAL EXPERIENCE

### AI Engineer - Docwyn.ai (USA-based startup, Remote)

May 2024 – July 2026

- Engineered Python/NLP document-intelligence workflows that converted unstructured business records into validated, query-ready outputs, improving extraction accuracy by 18%.
- Orchestrated Apache Airflow pipelines consolidating operational data from 120+ US retail locations into Snowflake for downstream AI, search and automation services.
- Optimized reusable Python/SQL processing components and Snowflake retrieval patterns, reducing data retrieval time by 40% across high-volume workflows.
- Automated document-processing and backend data workflows, improving consistency across AI-ready data preparation, reporting and product operations.

### AI Engineer Intern - Glowing Soft

Jul 2023 - Oct 2023

- Built Python ML/NLP pipelines processing 500K+ records per day, automating ingestion, cleaning and transformation while reducing manual processing time by 40%.
- Applied NLP classification and information extraction to contract datasets, converting free-form text into structured business fields and saving approximately 3 hours of manual review per day.
- Delivered 3 production-ready analytical/ML models and optimized SQL workloads on AWS RDS by 35%, accelerating experimentation and client-facing model outputs.

## PROJECTS

### AI Voice Customer Support Agent | LLMs, FastAPI, Twilio, ElevenLabs, PostgreSQL, Redis, Docker, Microservices

- Architected a real-time voice AI agent integrating telephony, speech-to-text, LLM tool calling, CRM/internal APIs and text-to-speech for context-aware customer support.
- Built modular FastAPI services for orchestration, session state, authentication and CRM/ticketing, backed by PostgreSQL/Redis and Dockerized for independent scaling.
- Implemented account/order lookup, ticket creation, conversation memory and human escalation with context handoff, connecting conversational AI to executable backend workflows.

### Multi-Agent Automation Platform with MCP Tooling | Python, MCP, FastAPI, Qdrant, PostgreSQL, Docker

- Designed Planner, Retrieval, Execution and Validation agents to decompose, route, execute and verify complex multi-step requests through specialized responsibilities.
- Built MCP client/server integrations exposing databases, vector search, document retrieval, files and REST services as discoverable agent tools.
- Implemented RAG with embeddings/Qdrant, structured outputs, persistent state, retries and validation gates in Dockerized FastAPI services for reliable agent execution.

## AWARDS

**Global Top 50 - NASA Space Apps Challenge 2024.** Ranked among the global top 50 from 57,000+ participants across 160+ countries. Built an AI-driven agriculture platform using NASA Earth-observation data, geospatial analytics and vision-language models for crop recommendations, yield insights, wildfire-risk alerts and sustainability guidance.

## PUBLICATIONS

**The Agreeable Swarm: Sycophantic Drift, Consensus Cascades, and Epistemic Collapse in Multi-Agent Large Language Model Systems (2026).** Research on multi-agent LLM reliability; introduces the ESRD/MA-ESRD framework and Sycophancy Amplification Factor (SAF), and analyzes consensus cascades, correlated-agent failure and mitigation strategies.

**When Small Models Create Big Overhead: The Specialization-Coordination Trade-off in Hybrid SLM-LLM Agent Systems (2026, manuscript).** Theoretical research on system-level efficiency in agentic AI; develops a coordination-cost framework, break-even analysis and architecture-selection framework for SLM, LLM, multi-SLM and hybrid agent systems.

## EDUCATION

Bachelor in Data Science - University of Central Punjab, Lahore